CLASS 7 — MATHS CHAPTER TEST SET B

Chapter 4: **Expressions Using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If x = 6, the value of 2x + 7 is: 1

(a) 19 **(b)** 15 **(c)** 26 **(d)** 21

2. If x = 3, the value of 7x + 4 is: 1

(a) 32 **(b)** 49 **(c)** 25 **(d)** 14

3. Simplify 7x + 2x − 4x. 1

(a) $5x^2$ **(b)** 10x **(c)** 5x **(d)** 13x

4. Simplify 5x + 5x − 3x. 1

(a) 22x **(b)** $7x^2$ **(c)** 13x **(d)** 7x

5. '3 times a number y' is written as: 1

(a) 3y **(b)** y/3 **(c)** y + 3 **(d)** y^3

6. The area of a square of side s is: 1

(a) 2s **(b)** s + s **(c)** 4s **(d)** s^2

7. The nth even number is: 1

(a) 2n **(b)** n + 2 **(c)** 2n + 1 **(d)** n^2

8. The expression 3a + 5a simplifies to: 1

(a) 35a **(b)** $8a^2$ **(c)** 15a **(d)** 8a

9. What is 5x − 3x? 1

(a) 8x **(b)** 15x **(c)** 2 **(d)** 2x

10. The number of terms in 3x + 2y − 5 is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What are like terms? Give an example. 2

12. Write the rule for the nth term of 4, 7, 10, 13 … 2

13. A notebook costs ₹p. Write the cost of 12 notebooks and of 12 notebooks and a ₹5 pen. 2

14. Expand 3(a + 4). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the perimeter of a rectangle with sides (3x + 2) and (2x + 1). 3

16. A square pattern uses 4n − (n − 1) sticks for n squares in a row. Simplify. 3

17. Ramu has ₹x. He spends ₹40 and gets ₹15 back. Write his money now and find it for x = 100. 3

OR

Show that 2(x + 4) and 2x + 8 give the same value for x = 5. 3

18. Three consecutive even numbers start at 2n. Write them and their sum. 3

OR

Subtract 3x − 2 from 7x + 5. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A phone plan costs ₹199 plus ₹3 per extra GB. 5

(i) Cost for g GB.
(ii) Cost for 6 GB.
(iii) GB used if bill is ₹250.
(iv) Write the cost for a family of 3 with g GB each.

OR

(i) Show that the sum of two consecutive numbers is odd. 5
(ii) Write the general form.
(iii) Find the sum for 17 and 18.

20. (i) Find the nth term of 2, 6, 10, 14 … 5
(ii) Find the 25th term.
(iii) Which term is 98?

OR

(i) Simplify 3(x + 4) − 2(x − 1) + 5. 5
(ii) Evaluate at x = 6.
(iii) Check by direct substitution.

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND